Ultra-Low-Noise Balanced Detectors for Optical Time-Domain Measurements

Qiming Lu, Qi Shen, Yuan Cao, Shengkai Liao, and Chengzhi Peng

***Abstract*— Two ultra-low-noise balanced detectors used for optical time-domain measurements are presented in this paper. Both simultaneously achieve high bandwidth and low noise that is extremely difficult to realize in commercial detectors. Two-stage amplification circuits based on transimpedance amplifiers (TIAs) were used for these detectors, with special modifications implemented for different applications. To further suppress noise, a low-noise junction field-effect transistor is connected between the photodiode and TIA to reduce the impact of amplifier leakage current in one of the detectors. Benefiting from this design, a 70-MHz sensitive detector with a gain of 3.2E5 V/W and low noise-equivalent power (NEP) density of 2.2 pW/rtHz was implemented. For another balanced detector used in higher bandwidth applications, the differential TIA circuit is used, with the detector achieving a 250-MHz bandwidth with a gain of 5E4 V/W, equivalent to NEP of 6.2 pW/rtHz. Due to the difficulty of achieving high bandwidth with low noise, we perform theoretical analysis and simulations for our designs to ensure that these two design goals are realized simultaneously. The performance of the detectors is consistent with our analysis. In addition, a simplified optical testing system was built to test and calculate the performance of the detectors in their respective applications. The results show that both detectors are well-balanced and achieve a common mode rejection ratio greater than 50 dB.**

***Index Terms*— Balanced detectors, circuit simulations, noise measurements, optoelectronic devices, time-domain measurements.**

I. Introduction

BALANCED detectors are widely used in time-domain measurements based on interference. To extract effective information from weak signal light, a local oscillator (LO) light is usually employed to interfere with it. The dc component in the output is suppressed by balanced detectors, and the remaining ac component is converted into electrical signals that contain the desired amplitude and phase information and then amplified. This kind of measurement is often referred to homodyne detection if the LO light is at the same frequency as the signal light, or heterodyne detection if it is at a different frequency [1]. The amplifier circuit designed with low noise and high gain enhances the sensitivity of the balanced detectors to a great extent. However, unlike photon counting based on single-photon detectors [2], [3], it is difficult to achieve photon number resolution using homodyne/heterodyne detection on balanced detectors due to the quantum noise in the signal light and the excess noise in the LO light [4], [5]. Therefore, balanced detectors are more suitable for applications that need to extract time-related information from weak signal light than count arriving photons in it, such as continuous-variable quantum key distribution (CVQKD) [6]–[10] and linear optical sampling (LOS) [11], [12].

A typical CVQKD experiment encodes random numbers into the amplitude and phase of a pulsed laser at quantum level at the transmitter and extracts time-related voltage information through homodyne detection at the receiver [7]. This method filters out the background light that is mixed with the signal light and ensures that it not only achieves high key distribution rate over short distances but also makes it possible to communicate in daylight [13]. According to [10], the common mode rejection ratio (CMRR) requirement is greater than 45 dB for a typical CVQKD experimental system; otherwise, the extra excess noise caused by the imbalance of detectors will decrease the key distribution rate. Although existing commercial balanced detectors are well established, they are not suitable for CVQKD because their CMRR are usually less than 40 dB [14], [15]. In addition, since the signal intensity at the receiver is typically only several photons per pulse, the electronic noise of the detectors must be suppressed in order to detect the weak signal light. In general, the electronic noise is at least 10 dB lower than shot noise [10], but commercial detectors usually do not pay attention to this performance. Therefore, we design a dedicated balanced detector and test the relevant performance for the experimental study of CVQKD in daylight.

Although there is sufficient research on balanced detectors used for CVQKD [6], [9], [10], [16], most of them do not achieve high bandwidth and low noise simultaneously. The detector in [6] possesses a shot noise to electronic noise ratio (noise ratio) of up to 14 dB. However, its bandwidth is only 1 MHz due to the long tail caused by the integral capacitor of the charge-sensitive amplifier. Using transimpedance amplifiers (TIAs) increase the bandwidth of detectors significantly, but other performance aspects are reduced with increasing noise [9], [10]. We find only a CMRR of 45 and 7.5 dB of noise ratio

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The authors are with the Hefei National Laboratory for Physical Sciences at the Microscale and Department of Modern Physics, University of Science and Technology of China, Hefei 230026, China, and also with the Chinese Academy of Sciences Center for Excellence and Synergetic Innovation Center in Quantum Information and Quantum Physics, University of Science and Technology of China, Shanghai 201315, China (e-mail: shenqi@ustc.edu.cn; skliao@ustc.edu.cn).

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in [24] in spite of achieving a bandwidth of 250 MHz. An ultra-high-noise ratio of 37 dB was demonstrated in [16], but it had a bandwidth of only a few megahertz. To satisfy the requirements of different CVQKD experiments, we take into account all aspects of performance when we design the low-noise balanced detector. A two-stage amplification circuit based on TIA is used to achieve high gain and bandwidth. A junction field-effect transistor (JFET) connected between the photodiode and TIA is used to interdict the loop of the amplifier leakage current flowing through the feedback resistor, so that the noise is suppressed [19]. Benefiting from these designs, the detector (CVQKD_BD) achieves a bandwidth of 70 MHz, with a gain of 3.2E5 V/W. The root-mean-square (RMS) value of the output voltage noise is about 6 mV, implying a noise-equivalent power (NEP) density of 2.2 pW/$\sqrt{\text{Hz}}$ that is lower than almost all commercial balanced detectors based on p-i-n photodiodes. The electronic noise is 16.8 dB lower than the shot noise at an LO power of 6.8 mW, better than most high-speed detectors in the existing literature, such as the 13 dB at 12 mW [9], 7.5 dB at 11.3 mW [24], or 13 dB at 6.4 mW [10] detectors.

LOS that performs high-precision time-domain sampling through heterodyne detection at high repetition rate femtosecond optical frequency combs is usually employed in spectrum analysis and time and frequency transfers [17], [18]. Since the repetition frequencies in LOS become higher than those in CVQKD, such as 200 MHz in [25], the design of the balanced detector proposed above is no longer applicable. Thus, a new balanced detector design with differential TIA circuit is proposed, and the JFET is removed so as to achieve high bandwidth as well. The high-speed balanced detector for LOS (LOS_BD) achieves a bandwidth of 250 MHz, with gain of 5E4 V/W. It has an output voltage noise RMS value of 4.9 mV, equivalent to NEP of 6.2 pW/$\sqrt{\text{Hz}}$. In our tests, the detector shows an ability to extract information from signal light at a minimum of 23 average photons per pulse, and further tests as well as theoretical analysis indicate the potential of detecting signal light at single-photon level.

In addition, we tested the balance performance of our detectors. The CMRR of CVQKD_BD and LOS_BD is 53 and 52 dB, respectively, nearly 20 dB higher than those of commercial detectors. This paper introduces the designs and tests of two low-noise balanced detectors for CVQKD and LOS. The remainder of this paper is structured as follows. Designs and simulations of balanced detectors are explained in Section II. We provide the experimental results in Section III. Comparison with other detectors and discussion of the performance of the detector are conducted in Section IV. The conclusions are presented in Section V.

II. Designs and Simulations

A. Schematic Designs

The simplified schematic diagrams of two balanced detectors are shown in Fig. 1(a) and (b). Two specially selected InGaAs p-i-n photodiodes (FGA01FC from Thorlabs, working at 1550 nm) were serially connected for photocurrent decrement. Operational amplifiers, OPA847, with an ultra-low equivalent input noise voltage density of 0.85 nV/$\sqrt{\text{Hz}}$ and 3.9 GHz gain-bandwidth product (GBP) from TI were used in these circuits. To prevent about 15-μA leakage current in the OPA847 from flowing through the transimpedance and constituting additional noise [19], a low-noise JFET (BF862) was added between the photodiodes and the TIA in the CVQKD_BD. The ultra-high input impedance of the BF862 suppresses the effects of leakage current on noise, equivalent to replacing the current noise source with a smaller JFET voltage noise source, about 0.8 nV/$\sqrt{\text{Hz}}$. However, the BF862 is not practical in high-speed applications because of its 8-pF junction capacitance, and the high repetition rate is critical for LOS applications to maintain the sampling frequency and precision. Therefore, we remove the JFET and replace the OPA847 with a differential operational amplifier, LTC6409 (Linear) that possesses a 10-GHz GBP and 3.3-V/ns differential slew rate (much more than 0.95 V/ns in OPA847) in LOS_BD. However, the 1.1- nV/$\sqrt{\text{Hz}}$ voltage noise source and unsubdued 70-μA bias current from LTC6409 will degrade the noise performance. In Section III-B, we discuss the bandwidth and noise performance under different conditions. Both detectors use OPA847 as second stage voltage amplifiers to achieve high gain. The two balanced detectors are made into four-layer printed circuit boards with the layout optimized, especially in routes from the photodiode to the TIA to reduce parasitic capacitance.

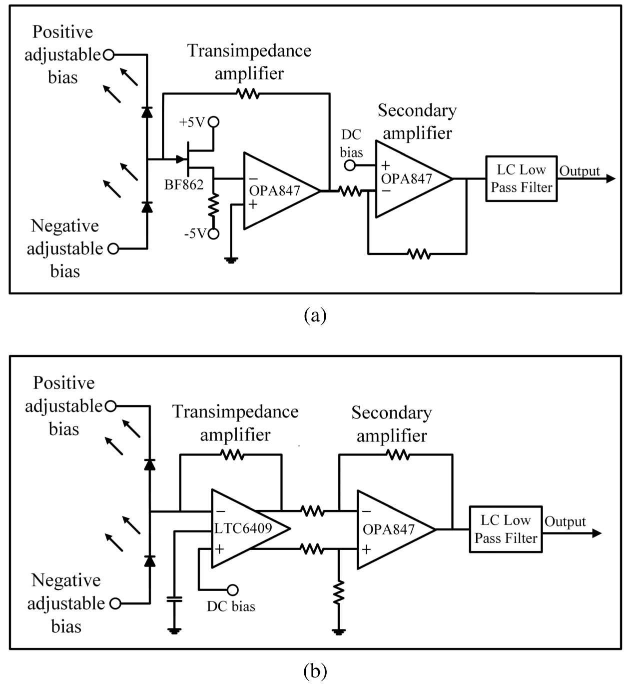

Fig. 1. Schematic of the detectors. (a) Schematic of CVQKD_BD. (b) Schematic of LOS_BD.

B. Theoretical Analysis and Simulations

Since the noise introduced by the voltage amplifier in the second stage of the detectors can easily be calculated using the noise figure (NF), we only analyze the noise of the TIA. Fig. 2 shows the simplified model of photovoltaic conversion and TIA for our balanced detectors, with the

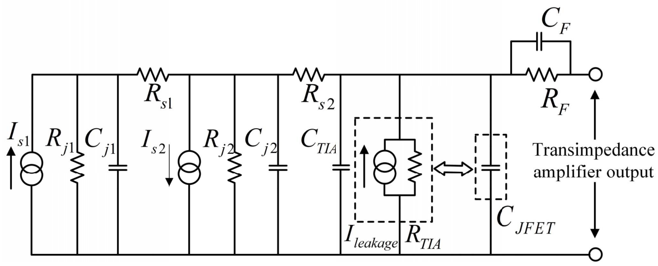

Fig. 2. Simplified model of photovoltaic conversion and TIA.

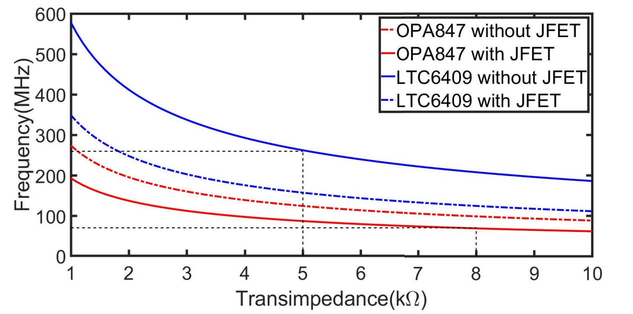

Fig. 3. Bandwidth simulations under different structures. The simulation parameters are shown in Table I.

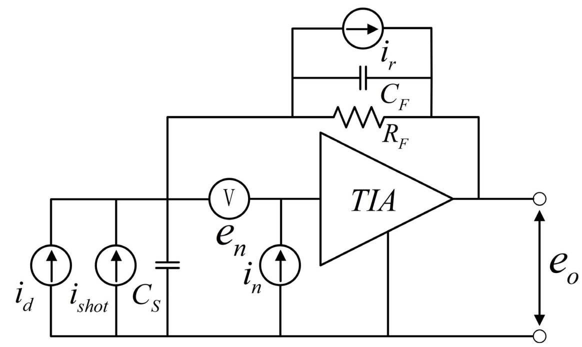

Fig. 4. Noise model of photoelectric conversion and TIA. Parallel resistances of amplifier and photodiodes are ignored because they are several orders of magnitude greater than the feedback resistance. The series equivalent resistances of the photodiodes are also ignored since they are only a few ohms.

model taken from [20]. The current sources I_{s1} and I_{s2} are reversed because of the current decrement structure. R_{j1} and R_{j2} represent the junction resistances of the photodiodes; these resistances are in the megaohm range and can be ignored for photodiodes operating in reverse-bias mode. The series resistances R_{s1} and R_{s2} are usually a few ohms, and for different photodiodes, these resistances will be slightly different. C_{j1} and C_{j2} represent the junction capacitances of the photodiodes, and for FGA01FC, these capacitances are 2 pF at 5 V under reverse-bias. There are three equivalent parts in TIA: input capacitance, C_{TIA}; input resistance, R_{TIA}; and leakage current, I_{leakage}. If a JFET is added before the amplifier, it will suppress the leakage current so that the two items behind can be ignored, but an additional C_{JFET} should be added because of its junction capacitance. To avoid self-oscillation of the amplifier, a feedback capacitance C_F must be used to maintain 45° phase margin for the circuit [21], and it is usually equal to

$$C_F = \sqrt{\frac{C_S}{2\pi R_F \mathrm{GBP}}} \tag{1}$$

where C_S is the sum of the capacitances mentioned above. To verify that the designs are reasonable, we perform theoretical analysis based on the above model.

1) Bandwidth and Gain: The bandwidth of detectors is mainly determined by the gain of the TIA, written as follows according to [20]:

$$f_{-3\,\mathrm{dB}} = \sqrt{\frac{\mathrm{GBP}}{2\pi R_F (C_S + C_F)}}. \tag{2}$$

R_F is the feedback resistance, and its value in CVQKD_BD is 8 kΩ. With a voltage amplification of 40, the total gain of CVQKD_BD is 3.2E5 V/W. There are some differences in LOS_BD that uses a differential amplifier to convert single-ended photocurrent to differential voltage. This makes the gain twice the transimpedance so that the total gain reaches 5E4 V/W with a transimpedance of 5 kΩ and voltage amplification of 5. Fig. 3 shows the simulation results for the bandwidth, and the solid lines represent our current designs. A balanced detector with OPA847 and JFET BF862 possesses a theoretical bandwidth of about 72 MHz, whereas the theoretical bandwidth in a detector with only LTC6409 is 260 MHz. The dashed lines show simulation results for the unimplemented designs, and the effect of junction capacitance in JFET on bandwidth can be observed by comparing the simulation results. The JFET will reduce bandwidth so it must be discarded in high-speed applications.

2) Noise: Fig. 4 shows the noise model for the photoelectric conversion and TIA. The relationship between the TIA noise and noise components is of the following form:

$$i^2_{\mathrm{TIA\_noise}} = i_d^2 + i^2_{\mathrm{shot}} + i_n^2 + i^2_{\mathrm{en}} + i_r^2. \tag{3}$$

i_d is the dark current in the photodiodes, with a value of 50 pA for FGA01FC at 5-V bias voltage. Since the value is very small compared with other noise components, we usually ignore it in noise calculations. i_{shot} represents the shot noise caused by the LO light; according to [20], the equivalent input noise current density of the shot noise caused by the photocurrent from the two photodiodes is given by

$$\begin{aligned} i_{\mathrm{shot}} &= \sqrt{i^2_{1\mathrm{shot}} + i^2_{2\mathrm{shot}}} = \sqrt{2qi_{1\mathrm{photo}} + 2qi_{2\mathrm{photo}x}} \\ &= \sqrt{2qR(\lambda)P_L}. \end{aligned} \tag{4}$$

In this formula, $R(\lambda)$ is the responsivity of the photodiodes and P_L refers to the power of the LO used for interference. i_n represents the equivalent input noise current source of the amplifier, and it is mainly caused by leakage current so that if a JFET is added, the i_n can be suppressed. i_{en} represents the noise caused by the equivalent input noise voltage source [21] of the amplifier, given by

$$i_{en} = \sqrt{\left(\frac{e_n}{R_F}\right)^2 + \frac{(2\pi e_n f_{-3\,\mathrm{dB}} C_S)^2}{3}}. \tag{5}$$

i_r is the thermal noise of the resistance, with the equivalent input noise current density given by

$$i_r = \sqrt{\frac{4kT}{R_F}}. \tag{6}$$

k represents Boltzmann's constant and T is the Kelvin temperature [22]. Substituting formulas (4)–(6) into formula (3), we obtain the total equivalent input noise current density as

$$i^2_{\text{TIA\_noise}} = \frac{4kT}{R_F} + 2qR(\lambda)P_l + i_n^2 + \left(\frac{e_n}{R_F}\right)^2 + \frac{(2\pi e_n f_{-3\,\text{dB}} C_S)^2}{3}. \tag{7}$$

Note that the above analysis only refers to noise of the TIA; the voltage amplifier must be taken into account to calculate the noise of the detectors. The NF refers to the signal-to-noise ratio (SNR) before and after the amplifier, given by the following equation [20]:

$$\begin{aligned}\text{NF} &= 10\lg\left(\frac{\text{SNR}_{\text{input}}}{\text{SNR}_{\text{output}}}\right) = 10\lg\left(\frac{\frac{P_i}{\text{Noise}_i}}{\frac{P_o}{\text{Noise}_o}}\right)\\ &= 20\lg\left(\frac{V_i}{V_o} * \frac{V_{\text{noise\_RMS}}}{V_{\text{noise\_TIA}}}\right) = 20\lg\left(\frac{1}{A_v} * \frac{V_{\text{noise\_RMS}}}{V_{\text{noise\_TIA}}}\right).\end{aligned} \tag{8}$$

A_v represents the gain of the voltage amplifier. According to formulas (7) and (8), if we regard the TIA noise density as a constant term across the entire bandwidth, the RMS of the total noise voltage can be estimated as shown in the following:

$$\begin{aligned} V_{\text{noise\_RMS}} &= 10^{\frac{\text{NF}}{20}} A_v V_{\text{TIA\_noise}} \\ &= 10^{\frac{\text{NF}}{20}} A_v R_F \sqrt{\int_0^{f_{-3\,\text{dB}}} i^2_{\text{noise}}\, df} \\ &= 10^{\frac{\text{NF}}{20}} A_v R_F \sqrt{f_{-3\,\text{dB}}}\, i_{\text{TIA\_noise}}. \end{aligned} \tag{9}$$

Commercial detectors typically use NEP density to evaluate the noise performance, and the value is given by [20]

$$\text{NEP} = \frac{V_{\text{noise\_RMS}}}{R_F A_v R(\lambda)\sqrt{f_{-3\,\text{dB}}}} = \frac{V_{\text{noise\_RMS}}}{GR(\lambda)\sqrt{f_{-3\,\text{dB}}}}. \tag{10}$$

G represents the total electronic gain of the detectors. Fig. 5 shows our noise simulation results for different detector designs, and these results show that the JFET can significantly reduce the RMS value of the noise voltage. The theoretical RMS noise voltage of the design using OPA847 with JFET is 8.3 mV at a transimpedance of 8 kΩ, signifying an NEP of 3.1 pW/$\sqrt{\text{Hz}}$. The same parameters have values of 7.3 mV at a transimpedance of 5 kΩ and 9 pW/$\sqrt{\text{Hz}}$ in designs that use LTC6409 without JFET. Comparing the simulation results of Figs. 3 and 5, it can be seen that it is difficult to balance both bandwidth and noise. Other optimizations such as controlling the printed circuit board traces to reduce parasitic inductance and filtering supply power multiple times to reduce ripple effects are helpful toward increasing the performance of the detectors.

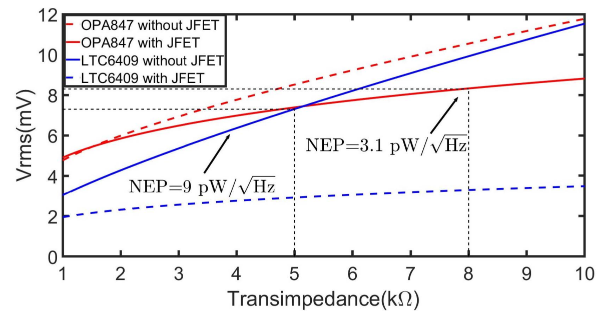

Fig. 5. Electronic noise simulations. The simulation parameters are shown in Table I.

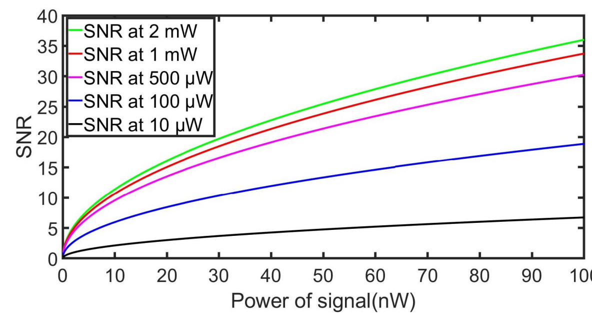

Fig. 6. SNR simulations for homodyne detection for LOS_BD. The simulation parameters are shown in Table I.

3) SNR of Homodyne Detection: The signal to electronic noise ratio is of great importance in the performance of LOS. According to [20], the shot-noise-limited SNR obtained in homodyne detection is twice as much as that obtained in heterodyne detection, meaning that we can measure the performance of LOS_BD through homodyne detection using only one femtosecond pulsed laser. The simplified theoretical output of the interference between LO and signal is given by

$$\begin{aligned} P_{\text{out}} &\propto |A_S e^{-i(\omega_S t + \phi_S)} + A_L e^{-i(\omega_L t + \phi_L)}|^2 \\ &\approx P_S + P_L + 2\sqrt{P_S P_L}\cos[(\omega_S - \omega_L)t + (\phi_S - \phi_L)]. \end{aligned} \tag{11}$$

A_S, ω_S, ϕ_S, P_S, and A_L, ω_L, ϕ_L,P_L are the carrier amplitude, frequency, phase, power of the signal, and LO, respectively. After decreasing the photocurrent using a balanced detector, the dc components: P_S and P_L are eliminated and the last ac component is retained. For homodyne detection, the output is given by

$$V_{\text{out}} = 2\text{GR}(\lambda)\sqrt{P_S P_L}. \tag{12}$$

We simulate the SNR of homodyne detection according to (9) and (12), and the results are shown in Fig. 6. The intensity of the signal light varies from 0.1 to 100 nW, and the five curves correspond to the SNR simulation results in different LOs. Increasing the intensity of the LO can improve the SNR, especially when the LO intensity is not so high. However, this effect decreases as the LO intensity increases so this method is not feasible at high powers. We verify this speculation later in subsequent tests.

TABLE I
SIMULATIONS PARAMETERS LIST

Condition	OPA847	OPA847&JFET	LTC6409	LTC6409&JFET
$C_S(pF)$	7.7	15.8	4.5	12.6
$e_n(nV/\sqrt{Hz})$	0.85	1.65	1.1	1.9
$i_n(pA/\sqrt{Hz})$	2.5	0	8.8	0
$R_F(k)$	8	8	5	5
$NF(dB)$	5	5	5	5
A_v	10	10	10	10

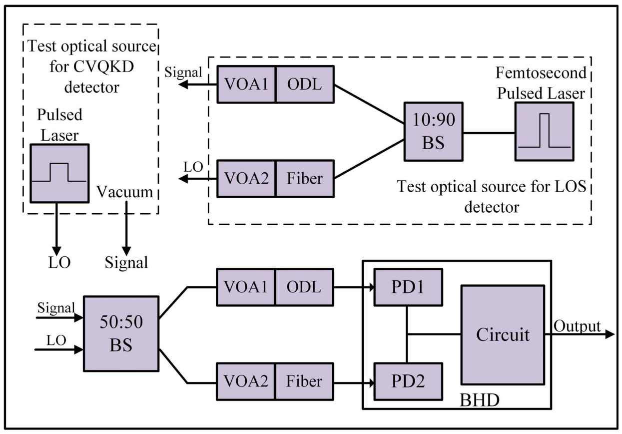

Fig. 7. Schematic for the optical path for testing the balanced detector.

III. TEST RESULTS

A. Test Block Diagram

We use the system shown in Fig. 7 to test our balanced detector. The signal light interferes with the LO on a 2 × 2 50:50 beam splitter (BS), with the outputs of the BS injected into the photodiodes of the balanced detector. Though the manual shows that the responsivity of the two photodiodes should be 1 A/W at 5 V bias, there are still slight differences between them. In addition, the optical path is slightly different between the two fibers from the BS to the balanced detector. Therefore, we must use variable optical attenuator (VOA, ranges from 0 to 40 dB) and optical delay line (ODL, MDL-002 from General Photonics) to precisely control the delay and attenuation of the two optical paths so as to achieve a high balanced test system. The dashed boxes in Fig. 7 are optical sources used to simulate different applications. The vacuum in the test optical source for the CVQKD detector refers to a situation when there is no signal light [10]. This means that only the LO is used for the CVQKD_BD test, which is a 1550-nm pulsed laser with 10% duty cycle and 40 MHz repetition rate. A 1550-nm femtosecond pulsed laser(ELMO Series, Menlo Systems) is used to simulate the optical frequency comb in the LOS. Its repetition rate is 100 MHz, and the pulsewidth is around 100 fs. This laser is split into two beams by a 10:90 BS, with the stronger part(90% of laser) used as LO and the weaker part (10% of laser) used as signal. VOAs in the dashed box are used to change the power of the femtosecond pulsed laser, whereas the ODL is used to adjust the signal overlaps with the LO to interfere.

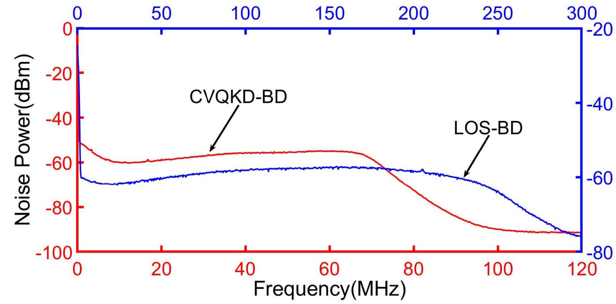

Fig. 8. Spectral tests for two detectors.

B. Bandwidth and Noise

In Fig. 8, we show the tests of balanced detectors on a spectrum analyzer, whereby each detector is followed by a corresponding low-pass filter. The bandwidth of the CVQKD_BD is about 70 MHz, and the RMS noise voltage is 6 mV, corresponding to an NEP of 2.2 pW/$\sqrt{\text{Hz}}$. The NEP of the LOS_BD is 6.2 pW/$\sqrt{\text{Hz}}$, with a 250-MHz bandwidth and RMS noise voltage of 4.9 mV. These test results are very close to our theoretical expectations, indicating that the model we have established is reasonable.

C. Common Mode Rejection Ratio Tests

Due to the difference in the frequency response of different photodiodes, the shape of electrical signals from the photodiodes will not be exactly the same even under similar input conditions. Thus, a 1550-nm pulsed laser with a 5-ns pulsewidth is used to compare the difference in response between different photodiodes in order to achieve high CMRR, and the closest two are selected as inputs to the balanced detectors. As the main reason for the difference in the frequency response is the difference in junction capacitances between the photodiodes [20] (which can be slightly compensated by the offset voltages applied at both ends), the bias voltages of the photodiodes in the detectors are designed to be tuned finely. Fig. 9 shows the CMRR test, obtained by comparing different responses of detectors to pulsed laser in the balanced and unbalanced modes [10]. The CMRR of the CVQKD_BD and LOS_BD is 53 and 52 dB, respectively. This will be helpful in suppressing the extra noise due to the imbalance of detectors.

D. Shot Noise to Electronic Noise Ratio Tests

For a CVQKD experiment, the electronic noise of the detector must be strictly controlled to improve the key distribution rate. According to [10], the electronic noise is at least 10 dB lower than the shot noise for a typical CVQKD experiment. In other words, the shot noise to electronic noise ratio should be greater than 10 dB. A good way for obtaining a higher noise ratio between the shot noise and electronic noise is to control the electronic noise and increase the gain [28]. We perform the shot noise to electronic noise ratio test for CVQKD_BD based on the method commonly used in CVQKD experiments [10]. The 1550-nm pulsed laser in Fig. 7 is modulated into a continuous laser of different powers and used as LO, whereas the power of the signal is zero, corresponding to the vacuum

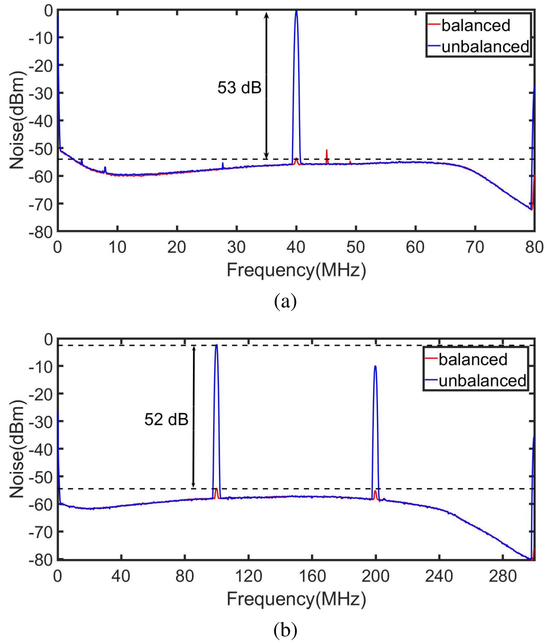

Fig. 9. CMRR tests for two detectors. In the balanced mode, two photodiodes of the detectors are injected with the same pulsed laser power, whereas only one photodiode is injected with pulsed laser in the unbalanced mode; the other is blocked. (a) CMRR test for CVQKD_BD. (b) CMRR test for LOS_BD.

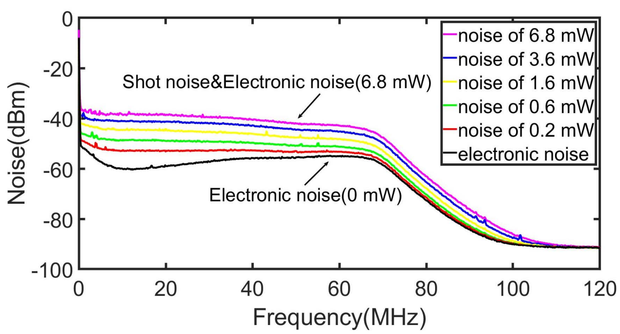

Fig. 10. Shot noise to electronic noise ratio test for CVQKD_BD.

state in CVQKD. Fig. 10 shows the electronic noise and shot noise levels of CVQKD_BD. Due to the ultra-low-noise design, we can find an average noise ratio greater than 10 dB at 1.6 mW and a maximum of 16.3 dB can be obtained at 6.8 mW over the 0–70-MHz bandwidth range.

E. Homodyne Detection Based on Femtosecond Pulsed Laser

Based on the previous discussion, we perform homodyne detections to test the performance of the balanced detector in LOS. The whole tests are based on the femtosecond pulsed laser we mentioned above. Since the bandwidth of commercial photodiodes is limited (1 GHz for FGA01FC), and it can be seen from Fig. 2 that the bandwidth limitation of photodiodes is equivalent to an RC low-pass filter, the actual responsivity of photodiodes is no longer 1 A/W. Our tests show that the actual responsivity of our photodiode for a femtosecond pulsed laser is 1.36E-5 A/W, 49 dB below the theoretical value.

We use a 100-μW LO and constantly adjust the signal power to test the dynamic range of the detector. A linear region

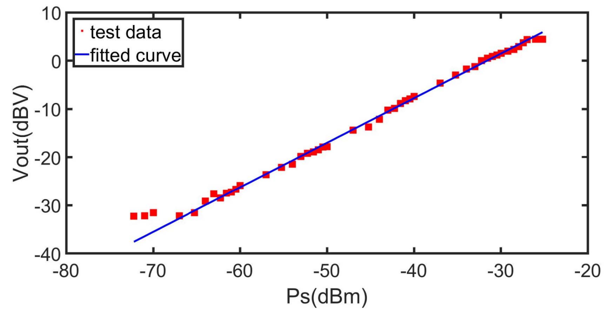

Fig. 11. Homodyne detection test for LOS_BD. Signal power is from 60 pW to 3 μW (−72.2 to −25.2 dBm).

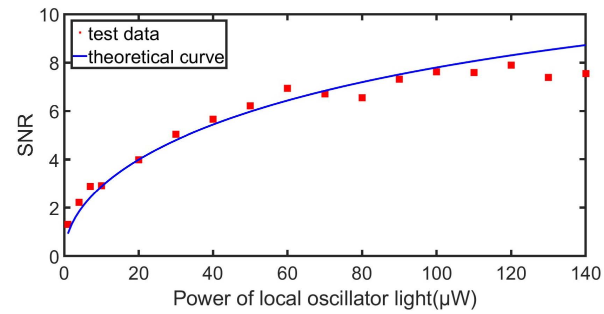

Fig. 12. Relationship between the power of the LO and SNR. The abscissa represents the power of the LO light from 1 to 140 μW, and the signal power is 10 nW. Note that the power here is the average power and must be converted to instantaneous power to achieve correct simulation. In addition, the responsivity of the photodiodes to femtosecond laser is not 1 A/W, so our simulation is based on the actual responsivity in the previous measurements.

of 37.8 dB from 0.3 to 1800 nW (−65.2 to −27.4 dBm) is measured on the basis of our test results in Fig. 11. The output of the detector is close to the electrical noise level when the power of signal light is 0.3 nW, thereby making each measurement similar at lower powers. When the power of the signal light exceeds 1800 nW, the amplifier of the detector begins to saturate. Since the output of the detector is proportional to $\sqrt{P_S P_L}$ according to formula (12), the linear trend is the normal operating range of LOS_BD, and we can estimate the linear region of signal light at other LO powers based on existing data.

Previous theoretical analysis indicates that it is useful to increase the power of the LO to achieve a high SNR when the LO power is weak. To verify this analysis, we perform the test in Fig. 12. The red points represent the measured data for the SNR. It can be seen that the SNR significantly improves with increasing powers of the LO in low power cases. The blue line represents the theoretical curve based on previous analysis, and we can obtain the same conclusion from it. Note that as the power increases, the SNR of LOS_BD begins to saturate faster than our theoretical analysis. This implies that the method of increasing LO power to improve SNR is more effective in low SNR condition than that in high SNR condition.

IV. Discussion

We achieved two ultra-low-noise balanced detectors with high bandwidth for CVQKD and LOS. Table II shows a comparison between our detectors and high-performance commercial detectors at 1550 nm. Compared with commercial

TABLE II
COMPARISON BETWEEN COMMERCIAL DETECTORS AND OUR DETECTORS

Detectors	Bandwidth (MHz)	Gain (V/W)	CMRR (dB)	NEP ($pW\sqrt{Hz}$)
C12668-04[14]	400	1E4	30	30
PDB425C[15]	75	2.5E5	35	5.2
BPD-1[26]	300	1E4	25	4
1817-FC[27]	80	4E4	25	3.5
CVQKD_BD	70	3.2E5	53	2.2
LOS_BD	250	5E4	52	6.2

TABLE III
COMPARISON BETWEEN BALANCED DETECTORS USED FOR CVQKD

Detectors	Ref.[6]	[9]	[10]	[16]	[24]	Ours
Wavelength(nm)	790	791	1550	1064	1064	1550
Bandwidth(MHz)	1	100	100	5	250	70
CMRR(dB)	85	52.4	46	75	45	53
Noise ratio(dB)	14	13	13	37	7.5	16.8

detectors, CVQKD_BD possesses the lowest NEP. The noise performance of LOS_BD is slightly worse than that of CVQKD_BD due to the elimination of JFET. However, LOS_BD achieves low noise while maintaining higher gain and CMRR than commercial detectors, and this makes it perform better in high-speed applications. The following represents a discussion of the respective applications.

A. Discussion for CVQKD_BD

Due to the influence of pulse trailing, the key distribution rate will decrease when the pulse repetition rate of the experiment increases and approaches the bandwidth of the balanced detector. The optimal repetition rate is generally one-third of the bandwidth [10], implying that our detector can support a 23-MHz repetition rate CVQKD experiment. Table III shows a comparison between our detector with the existing balanced detector research for CVQKD, and it can be seen that it is difficult to increase the bandwidth, CMRR, and shot noise to electronic noise ratio simultaneously. The methods adopted for increasing the bandwidth are often in contradiction with those adopted for increasing the CMRR and noise ratio. Our detector takes the bandwidth into account while achieving ultra-low-noise that will be more versatile to CVQKD experiments in different channel environments.

B. Discussion for LOS_BD

SNR is a key factor in LOS, and the test above shows that our detector can resolve a minimum signal power of 0.3 nW (equivalent to 23 photons per pulse) at 100-μW LO. Further tests and theoretical calculations indicate that we can improve the SNR by increasing the power of the LO. Since the SNR is proportional to the square root of the power of the LO at low SNR conditions, we may extract information from an average of two photons per pulse signal light by increasing the power of the LO to 1 mW, or from an average of one photon per pulse signal light by increasing the power of the LO to 2 mW. However, it is difficult to increase the SNR when the power of the LO continues to increase, according to Figs. 6 and 12. On the basis of our previous analysis, the capacitance before the TIA significantly reduces the bandwidth and increases extra noise. Therefore, using smaller junction capacitance photodiodes and optimizing the traces at the input section of the amplifier to reduce parasitic effects will likely improve the performance of the detector.

V. CONCLUSION

We verified a noise model for balanced detectors using TIA circuit followed by a voltage amplifier circuit. Two low-noise balanced detectors for CVQKD and LOS were realized on the basis of the existing analysis. The CVQKD_BD is well-balanced at all performance aspects, and the bandwidth is 70 MHz with a gain of 3.2E5 V/W and an NEP of 2.2 $pW/\sqrt{Hz}$. The JFET is removed in LOS_BD to guarantee high bandwidth, and the detector eventually reaches 250 MHz bandwidth with a gain of 5E4 V/W, and an NEP of 6.2 $pW/\sqrt{Hz}$. Both of them achieved a CMRR greater than 50 dB. The ultra-low-noise performance makes the electronic noise 16.8 dB lower than shot noise in CVQKD_BD that is better than most value recorded for high-speed detectors in the existing literature. This performance may help LOS_BD achieve the ability to extract information from a minimum of 23 average photons per pulse signal light at a power of 100 μW for the LO.

REFERENCES

[1] S. Machida and Y. Yamamoto, "Quantum-limited operation of balanced mixer homodyne and heterodyne receivers," *IEEE J. Quantum Electron.*, vol. QE-22, no. 5, pp. 617–624, May 1986.

[2] A. Lacaita, P. A. Francese, F. Zappa, and S. Cova, "Single-photon detection beyond 1 μm: Performance of commercially available germanium photodiodes," *Appl. Opt.*, vol. 33, no. 30, pp. 6902–6918, Oct. 1994.

[3] A. Lacaita, F. Zappa, S. Cova, and P. Lovati, "Single-photon detection beyond 1 μm: Performance of commercially available InGaAs/InP detectors," *Appl. Opt.*, vol. 35, no. 16, pp. 2986–2996, Jun. 1996.

[4] G. L. Abbas, V. W. S. Chan, and T. K. Yee, "Local-oscillator excess-noise suppression for homodyne and heterodyne detection," *Opt. Lett.*, vol. 8, no. 8, pp. 419–421, Aug. 1983.

[5] J. Shapiro, "Quantum noise and excess noise in optical homodyne and heterodyne receivers," *IEEE J. Quantum Electron.*, vol. 21, no. 3, pp. 237–250, Mar. 1985.

[6] H. Hansen *et al.*, "Ultrasensitive pulsed, balanced homodyne detector: Application to time-domain quantum measurements," *Opt. Lett.*, vol. 26, no. 21, pp. 1714–1716, Nov. 2001.

[7] F. Grosshans, G. Van Assche, J. Wenger, R. Brouri, N. J. Cerf, and P. Grangier, "Quantum key distribution using Gaussian-modulated coherent states," *Nature*, no. 421, pp. 238–241, Jan. 2003.

[8] J. Lodewyck *et al.*, "Quantum key distribution over 25 km with an all-fiber continuous-variable system," *Phys. Rev. A, Gen. Phys.*, vol. 76, no. 4, pp. 2305–2315, Oct. 2007.

[9] R. Kumar, E. Barrios, A. MacRae, E. Cairns, E. H. Huntington, and A. I. Lvovsky, "Versatile wideband balanced detector for quantum optical homodyne tomography," *Opt. Commun.*, vol. 285, no. 24, pp. 5259–5267, Nov. 2012.

[10] Y.-M. Chi *et al.*, "A balanced homodyne detector for high-rate Gaussian-modulated coherent-state quantum key distribution," *New J. Phys.*, vol. 13, no. 1, p. 013003, Jan. 2011.

[11] C. Dorrer, D. C. Kilper, H. R. Stuart, G. Raybon, and M. G. Raymer, "Linear optical sampling," *IEEE Photon. Technol. Lett.*, vol. 15, no. 12, pp. 1746–1748, Dec. 2003.

[12] I. Coddington, W. C. Swann, and N. R. Newbury, "Coherent linear optical sampling at 15 bits of resolution," *Opt. Lett.*, vol. 34, no. 14, pp. 2153–2155, Jul. 2009.

[13] B. Heim *et al.*, "Atmospheric continuous-variable quantum communication," *New J. Phys.*, vol. 16, no. 11, p. 113018, Nov. 2014.

[14] Hamamatsu Photonics Co., Ltd. Tokyo, Japan. *Balanced Detectors Index*. Accessed: Dec. 25, 2018. [Online]. Available: https://www.hamamatsu.com/us/en/product/optical-sensors/photodiodes/balanced-detectors/index.html

[15] Thorlabs, Inc. Newton, NJ, USA. *Balanced Detectors Index*. Accessed: Dec. 25, 2018. [Online]. Available: https://www.thorlabs.com/navigation.cfm?guide_id=2120

[16] X. Jin, J. Su, Y. Zheng, C. Chen, W. Wang, and K. Peng, "Balanced homodyne detection with high common mode rejection ratio based on parameter compensation of two arbitrary photodiodes," *Opt. Express*, vol. 23, no. 18, pp. 23859–23866, Sep. 2015.

[17] I. Coddington, W. C. Swann, and N. R. Newbury, "Coherent dual-comb spectroscopy at high signal-to-noise ratio," *Phys. Rev. A, Gen. Phys.*, vol. 82, no. 4, pp. 3817–3830, Oct. 2010.

[18] F. R. Giorgetta, W. C. Swann, L. C. Sinclair, E. Baumann, I. Coddington, and N. R. Newbury, "Optical two-way time and frequency transfer over free space," *Nature Photon.*, vol. 7, pp. 434–438, Apr. 2013.

[19] D. Huang, J. Fang, C. Wang, P. Huang, and G.-H. Zeng, "A 300-MHz bandwidth balanced homodyne detector for continuous variable quantum key distribution," *Chin. Phys. Lett.*, vol. 30, no. 11, p. 114209, Nov. 2013.

[20] S. B. Alexander, *Optical Communication Receiver Design*. Bellingham, WA, USA: SPIE, 1997, pp. 121–216.

[21] L. Orozco. (May 2013). *Programmable-Gain Transimpedance Amplifiers Maximize Dynamic Range in Spectroscopy Systems*. [Online]. Available: http://www.analog.com/en/analog-dialogue/articles/programmable-gain-transimpedance-amplifiers.html

[22] H. Nyquist, "Thermal agitation of electric charge in conductors," *Phys. Rev.*, vol. 32, no. 1, pp. 110–113, Jul. 1928.

[23] X. Ramus. (Nov. 2009). *Transimpedance Considerations for High-Speed Amplifiers*. [Online]. Available: http://www.ti.com/lit/an/sboa122/sboa122.pdf

[24] R. Okubo, M. Hirano, Y. Zhang, and T. Hirano, "Pulse-resolved measurement of quadrature phase amplitudes of squeezed pulse trains at a repetition rate of 76 MHz," *Opt. Lett.*, vol. 33, no. 13, pp. 1458–1460, Jul. 2008.

[25] J. D. Deschênes *et al.*, "Synchronization of distant optical clocks at the femtosecond level," *Phys. Rev. X*, vol. 6, no. 2, pp. 021016–021031, Apr./May 2016.

[26] Insight Photonic Solutions, Inc. Lafayette, CO, USA. *High Performance Balanced Photodetector (BPD-1)*. Accessed: Dec. 25, 2018. [Online]. Available: http://www.sweptlaser.com/content/high-performance-balanced-photodetector-bpd-1-0

[27] Newport Corporation. Irvine, CA, USA. *80 MHz Balanced Optical Receivers*. Accessed: Dec. 25, 2018. [Online]. Available: https://www.newport.com.cn/f/80-mhz-balanced-photoreceivers

[28] D. Huang, P. Huang, D. Lin, and G. Zeng, "Long-distance continuous-variable quantum key distribution by controlling excess noise," *Sci. Rep.*, vol. 6, Jan. 2016, Art. no. 19201.